

Finite-rank aDP operators are lush

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Source and target

The source paper is Kadets, Martín, Merí, and Pérez, *Spear operators between Banach spaces*, arXiv:1701.02977. All Banach spaces below are real or complex.

After Proposition 6.5, the source paper says that it is not known whether the finite-dimensional codomain theorem, or part of it, remains true when only the range of the operator is finite-dimensional. Chapter 10 also asks whether the rank-one results extend to finite-rank operators. The result below proves the aDP/spear/lush equivalence part for finite-dimensional range, hence for finite-rank operators.

Definitions and source facts

A norm-one operator $G : X \rightarrow Y$ has the alternative Daugavet property (aDP) if the alternative Daugavet equation holds for every rank-one operator $T : X \rightarrow Y$. A spear operator satisfies the analogous equation for every bounded operator $T : X \rightarrow Y$, and every lush operator is spear.

We use two facts from the source paper.

1. Theorem 3.2(iv): if $G : X \rightarrow Y$ has the aDP, then for every $x^* \in X^*$ the set

$$D(x^*) = \{y^* \in \text{ext } B_{Y^*} : \|G^*y^* + \mathbb{T}x^*\| = 1 + \|x^*\|\}$$

is dense G_δ in $(\text{ext } B_{Y^*}, w^*)$.

2. Proposition 3.32(a): if

$$\{y^* \in B_{Y^*} : G^*y^* \in \text{Spear}(X^*)\}$$

is norming for Y , then G is lush.

Proof Intuition

For one test vector $x^* \in X^*$, the aDP gives many extreme functionals y^* for which G^*y^* satisfies the spear equation against that particular x^* . For finitely many test vectors, Baire category gives one y^* satisfying all finitely many equations inside any desired weak-star slice.

The finite-rank assumption lets us pass from finitely many equations to all equations. Indeed, the set $G^*(B_{Y^*})$ sits in a finite-dimensional subspace of X^* , so weak-star compact slices of B_{Y^*} have norm-compact images under G^* . The closed sets expressing each spear equation have the finite-intersection property, hence a common point. This common point is a spear functional in X^* , and because the weak-star slice was arbitrary, such functionals norm Y . Proposition 3.32(a) then turns this norming family into lushness of G .

Main theorem

Theorem 1. *Let X, Y be Banach spaces and let $G : X \rightarrow Y$ be a norm-one finite-rank operator. If G has the alternative Daugavet property, then G is lush.*

Proof. It suffices, by Proposition 3.32(a) of the source paper, to prove that

$$\mathcal{N} = \{y^* \in B_{Y^*} : G^*y^* \in \text{Spear}(X^*)\}$$

is norming for Y .

Fix $y_0 \in S_Y$ and $0 < \varepsilon < 1$. Let

$$U = \{y^* \in B_{Y^*} : \text{Re } y^*(y_0) > 1 - \varepsilon\}.$$

The set $U \cap \text{ext } B_{Y^*}$ is nonempty. Indeed, U is a nonempty weak-star slice of B_{Y^*} , and if every extreme point of B_{Y^*} lay outside U , then the weak-star closed convex hull of the extreme points would also lie outside U , contradicting Krein–Milman.

For each $x^* \in X^*$, Theorem 3.2(iv) gives a dense G_δ subset

$$D(x^*) = \{y^* \in \text{ext } B_{Y^*} : \|G^*y^* + \mathbb{T}x^*\| = 1 + \|x^*\|\}$$

of $(\text{ext } B_{Y^*}, w^*)$. Hence, if $F \subset X^*$ is finite, then

$$U \cap \text{ext } B_{Y^*} \cap \bigcap_{x^* \in F} D(x^*) \neq \emptyset.$$

Now put $E = G^*(Y^*)$. Since G has finite rank, E is a finite-dimensional subspace of X^* . Let

$$K = G^*(\overline{U}^{w^*}) \subset E.$$

This set is compact in norm, because $\overline{U}^{w^*}$ is weak-star compact and E is finite-dimensional. For each $x^* \in X^*$, define

$$K_{x^*} = \{e \in K : \|e + \mathbb{T}x^*\| = 1 + \|x^*\|\}.$$

Each K_{x^*} is closed in K . The preceding Baire-category observation says that the family $\{K_{x^*} : x^* \in X^*\}$ has the finite-intersection property: for finitely many $x_1^*, \dots, x_n^*$, pick $y^* \in U \cap \text{ext } B_{Y^*} \cap \bigcap_{j=1}^n D(x_j^*)$, and then $G^*y^* \in \bigcap_{j=1}^n K_{x_j^*}$.

By compactness of K , there exists $e_0 \in \bigcap_{x^* \in X^*} K_{x^*}$. Since $e_0 \in K = G^*(\overline{U}^{w^*})$, choose $y_0^* \in \overline{U}^{w^*}$ with $G^*y_0^* = e_0$. The equality for $x^* = 0$ gives $\|e_0\| = 1$, and the equality for every $x^* \in X^*$ gives

$$\|e_0 + \mathbb{T}x^*\| = 1 + \|x^*\| \quad (x^* \in X^*).$$

Thus $e_0 = G^*y_0^* \in \text{Spear}(X^*)$. Also $\text{Re } y_0^*(y_0) \geq 1 - \varepsilon$, because $y_0^* \in \overline{U}^{w^*}$. Therefore $\mathcal{N}$ is norming for Y . Proposition 3.32(a) now implies that G is lush. $\square$

Corollary 1. *Let $G : X \rightarrow Y$ be a norm-one finite-rank operator. Then the following are equivalent:*

$$G \text{ is lush}, \quad G \text{ is a spear operator}, \quad G \text{ has the aDP}.$$

Proof. Every lush operator is spear, and every spear operator has the aDP. The theorem proves the converse implication from aDP to lush for finite-rank operators. $\square$

Consequences and limitations

This theorem strengthens the earlier lane-19 norm-attainment packet: finite-rank aDP operators attain their norm because they are lush, and the source paper proves that lush operators attain their norm.

The theorem does not settle the finite-rank analogue of Proposition 7.8 of the source paper. In other words, it remains open from this argument whether $G^* : Y^* \rightarrow X^*$ must be lush whenever $G : X \rightarrow Y$ is finite-rank and has the aDP.

Novelty notes

The proof uses only source-paper tools plus finite-dimensional compactness of $G^*(B_{Y^*})$. Cheap run-index searches found no prior packet for this finite-rank aDP/lushness equivalence. Local checks against arXiv:2306.02645 and arXiv:2306.02638 did not find the statement. Web searches on 2026-06-16 for exact combinations of “finite-rank”, “alternative Daugavet property”, “finite-dimensional range”, “spear operators”, and “lush” found no exact duplicate.

References

- [1] V. Kadets, M. Martín, J. Merí, and D. Pérez, *Spear operators between Banach spaces*, arXiv:1701.02977.